

ECON 320 – Practice Problem Set 1

Sets and Functions

Mathematical Foundations for Economic Analysis

Course: ECON 320: Introduction to Mathematical Economics

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1 Practice Questions

1.1 Section A: Sets and Set Operations

A1. (Set Builder Notation) Write each of the following sets in set-builder notation and determine whether it is finite or infinite:

- (a) The set of all even natural numbers less than 20.
- (b) The set of all real numbers whose square is less than 16.
- (c) The set of all prices in $[0, 100]$ that make demand positive, given $D(P) = 50 - P$.

A2. (Set Operations) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$, and $C = \{5, 6, 7, 8, 9\}$. Compute:

- (a) $A \cup B$ (b) $A \cap B$ (c) $A \setminus B$ (d) $B \setminus A$ (e) $(A \cup B)^c$ (f) $A \cap B \cap C$
(g) $A \cup (B \cap C)$

A3. (Power Set) Find the power set of $A = \{x, y\}$. How many elements does $\mathcal{P}(\mathcal{P}(A))$ have?

A4. (Cartesian Product) Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. List all elements of $A \times B$ and $B \times A$. Are they equal?

A5. (De Morgan's Laws) Verify De Morgan's Law $(A \cup B)^c = A^c \cap B^c$ using $U = \{1, \dots, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$.

A6. (Economic Application) A consumer has income $w = 100$ and faces prices $p_1 = 2$, $p_2 = 5$. Define:

$$A = \{(x_1, x_2) \in \mathbb{R}_+^2 : 2x_1 + 5x_2 \leq 100\}$$

$$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 \geq 10\}$$

Describe $A \cap B$ geometrically. What does this set represent economically?

A7. (Proof) Prove the distributive law: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ using the element-chasing method.

A8. (Supremum and Infimum) For each set, find $\sup$, $\inf$, and state whether $\max$ and $\min$ exist:

- (a) $A = \{\frac{n}{n+1} : n \in \mathbb{N}\}$
- (b) $B = (-3, 5]$
- (c) $C = \{x \in \mathbb{R} : x^2 < 9\}$
- (d) $D = \{x \in \mathbb{R} : x^3 \leq 8\}$

1.2 Section B: Functions

B1. (Domain and Range) Find the natural domain and range of:

(a) $f(x) = \sqrt{9 - x^2}$

(b) $g(x) = \frac{x - 1}{x^2 - 4}$

(c) $h(x) = \ln(x^2 - 1)$

(d) $k(x, y) = \sqrt{x} + \sqrt{y}$ (treat as a function from $\mathbb{R}^2$)

B2. (Injectivity and Surjectivity) For each function, determine whether it is injective, surjective, bijective, or none:

(a) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x - 7$

(b) $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = x^3 - x$

(c) $h : \mathbb{R} \rightarrow \mathbb{R}^+, h(x) = e^x$

(d) $k : [0, \pi] \rightarrow [-1, 1], k(x) = \cos x$

(e) $\ell : \mathbb{R}_+ \rightarrow \mathbb{R}_+, \ell(x) = x^3$

B3. (Inverse Functions) Find the inverse of each function and state its domain:

(a) $f(x) = 5x + 3$

(b) $g(x) = \frac{x + 1}{x - 2}, x \neq 2$

(c) $h(x) = e^{2x-1}$

(d) $k(x) = x^3 + 2$ on $\mathbb{R}$

B4. (Composition) Let $f(x) = 2x^2 - 1$ and $g(x) = \sqrt{x + 3}$ (defined on $[-3, \infty)$). Compute:

(a) $(g \circ f)(x)$ and state its domain.

(b) $(f \circ g)(x)$ and state its domain.

(c) $(g \circ f)(2)$ and $(f \circ g)(1)$.

(d) Are $g \circ f$ and $f \circ g$ equal? What does this illustrate?

B5. (Pre-image) Let $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 - 4$. Find:

(a) $f^{-1}(\{0\})$ (b) $f^{-1}([-4, 0])$ (c) $f^{-1}((0, 5])$ (d) $f^{-1}(\{-5\})$

B6. (Homogeneity) Determine whether each function is homogeneous, and if so, find the degree:

(a) $f(K, L) = 3K^{0.4}L^{0.6}$

(b) $g(x, y) = x^2 + xy + y^2$

(c) $h(x, y) = \frac{x^2y}{x^3 + y^3}$

(d) $k(x, y) = x + y + 1$

B7. (Concavity and Convexity) Determine whether each function is convex, concave, or neither:

(a) $f(x) = x^4$ on $\mathbb{R}$

(b) $g(x) = x^3$ on $\mathbb{R}$

(c) $h(x) = -2x^2 + 4x - 1$ on $\mathbb{R}$

(d) $k(x) = |x|$ on $\mathbb{R}$

B8. (Economic Application — Full Question) A monopolist faces inverse demand $P(Q) = 100 - 2Q$ and has total cost $C(Q) = Q^2 + 10Q + 50$.

(a) Define the profit function $\pi(Q)$ and state its domain.

(b) Find the range of π .

(c) Is π concave? Justify.

(d) Find the profit-maximising quantity and maximum profit.

(e) Find $\pi^{-1}(\{0\})$ (the break-even quantities) and interpret.